

Solved selected problems of Introductory functional analysis with applications - Erwin Kreyszig

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Chapter 3 - Inner Product Spaces. Hilbert Spaces

3.1 - Inner Product Space. Hilbert Space

Proof. 1 Let us prove (4) as follows

$$\begin{aligned}\|x + y\|^2 + \|x - y\|^2 &= \langle x + y, x + y \rangle + \langle x - y, x - y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle + \langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle \\ &= 2\langle x, x \rangle + 2\langle y, y \rangle \\ &= 2(\|x\|^2 + \|y\|^2)\end{aligned}$$

Where we used the definition of the norm $\|x\| = \sqrt{\langle x, x \rangle}$ and the distributive property of the inner products. $\square$

Proof. 2 Let $x \perp y$ in an inner product space X then

$$\begin{aligned}\|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \langle x, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + \|y\|^2\end{aligned}$$

Where we used that $\langle x, y \rangle = \langle y, x \rangle = 0$ since $x \perp y$.

Suppose now we have m mutually orthogonal vectors, i.e. $x_1 \perp x_2 \perp \dots \perp x_m$ then

$$\begin{aligned}\|x_1 + x_2 + \dots + x_m\|^2 &= \langle x_1 + x_2 + \dots + x_m, x_1 + x_2 + \dots + x_m \rangle \\ &= \langle x_1, x_1 \rangle + \langle x_1, x_2 \rangle + \dots + \langle x_m, x_{m-1} \rangle + \langle x_m, x_m \rangle \\ &= \langle x_1, x_1 \rangle + \langle x_2, x_2 \rangle + \dots + \langle x_m, x_m \rangle \\ &= \|x_1\|^2 + \|x_2\|^2 + \dots + \|x_m\|^2\end{aligned}$$

Where we used that $\langle x_i, x_j \rangle = 0$ for $i \neq j$ since they are orthogonal. $\square$

Proof. **3** Let $x, y \in \mathbb{R}$ and suppose we have the following property

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2$$

then since $\|x\|^2 = \langle x, x \rangle$ above equation implies that

$$\begin{aligned}\langle x + y, x + y \rangle &= \langle x, x \rangle + \langle y, y \rangle \\ \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle &= \langle x, x \rangle + \langle y, y \rangle \\ \langle x, y \rangle + \langle y, x \rangle &= 0\end{aligned}$$

But since $x, y \in \mathbb{R}$ we have that $\langle x, y \rangle = \langle y, x \rangle$, hence

$$\langle x, y \rangle = \langle y, x \rangle = 0$$

Therefore $x \perp y$.

Let now $x, y \in \mathbb{C}$ and suppose that the property holds in this case too. Then as before must be that

$$\langle x, y \rangle + \langle y, x \rangle = 0$$

Also, we know that $\langle y, x \rangle = \overline{\langle x, y \rangle}$ hence we need that

$$\langle x, y \rangle = -\overline{\langle x, y \rangle}$$

But since we are assuming a complex inner product we have that $\langle x, y \rangle$ is of the form $a + bi$ hence

$$a + bi = -(a - bi) = -a + bi$$

Equating the real and imaginary parts we see that $a = 0$ and $b = b$. So for example if $x = 1 + i$ and $y = 1 - i$ we see that the Pythagorean property holds, as we show below

$$\|1 + i + 1 - i\|^2 = \|2\|^2 = 4 = 2 + 2 = \|1 + i\|^2 + \|1 - i\|^2$$

But inner product gives us

$$\langle x, y \rangle = (1 + i)\overline{(1 - i)} = (1 + i)(1 + i) = 1 + i + i - 1 = 2i$$

Therefore x and y are not orthogonal. □

Proof. **6** Let $x \neq 0$ and $y \neq 0$

- (a) Let $x \perp y$ we want to prove that $\{x, y\}$ is a linearly independent set i.e. we want to prove

$$\alpha_1 x + \alpha_2 y = 0$$

Only for $\alpha_1 = \alpha_2 = 0$. Suppose the opposite we want to arrive at a contradiction, let $\alpha_1 \neq 0$ and $\alpha_2 = 0$ then $\alpha_1 x + 0y = 0$ i.e. $x = 0$ but we said that $x \neq 0$ so it must be that $\alpha_1 \neq 0$ and $\alpha_2 \neq 0$. Then we can write that

$$x = -\frac{\alpha_2}{\alpha_1}y$$

Also, we know that $\langle x, y \rangle = 0$ since $x \perp y$ then

$$\langle x, y \rangle = \langle -\frac{\alpha_2}{\alpha_1}y, y \rangle = -\frac{\alpha_2}{\alpha_1} \langle y, y \rangle = 0$$

But this implies that $y = 0$ and we said that $y \neq 0$, a contradiction.

Therefore must be that $\{x, y\}$ is a linearly independent set.

- (b) Now, let $\{x_1, \dots, x_m\}$ be a set of mutually orthogonal non-zero vectors, we want to show they are a linearly independent set.

As before, suppose the opposite, we want to arrive at a contradiction.

If $\alpha_i \neq 0$ for only the i th vector then $x_i = 0$ and that cannot happen.

So let us assume that $\alpha_i \neq 0$ for $1 \leq i \leq m$, if not all α_i s were different from zero the following still applies, then

$$\begin{aligned} \alpha_1 x_1 + \dots + \alpha_n x_n &= 0 \\ x_1 &= -\frac{\alpha_2}{\alpha_1}x_2 - \dots - \frac{\alpha_n}{\alpha_1}x_n \end{aligned}$$

Also we know that $\langle x_1, x_2 \rangle = 0$ then

$$\begin{aligned} \langle x_1, x_2 \rangle &= \langle -\frac{\alpha_2}{\alpha_1}x_2 - \dots - \frac{\alpha_n}{\alpha_1}x_n, x_2 \rangle \\ &= -\frac{\alpha_2}{\alpha_1} \langle x_2, x_2 \rangle - \dots - \frac{\alpha_n}{\alpha_1} \langle x_n, x_2 \rangle \\ &= -\frac{\alpha_2}{\alpha_1} \langle x_2, x_2 \rangle \\ &= 0 \end{aligned}$$

Where we used that $\langle x_i, x_2 \rangle = 0$ for $1 \leq i \leq m$ and $i \neq 2$. But the last equality implies that $x_2 = 0$ which is a contradiction.

Therefore must be that the orthogonal nonzero vectors $\{x_1, \dots, x_m\}$ are a linearly independent set.

□

Proof. 7 Let $\langle x, u \rangle = \langle x, v \rangle$ for all x , then must also be true for $x = u$ i.e. $\langle u, u \rangle = \langle u, v \rangle$ and for $x = v$ i.e. $\langle v, u \rangle = \langle v, v \rangle$.
Now, let us compute $\langle u - v, u - v \rangle$ as follows

$$\begin{aligned}\langle u - v, u - v \rangle &= \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle \\ &= \langle u, v \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, u \rangle \\ &= 0\end{aligned}$$

But then this implies that $u - v = 0$. Therefore $u = v$. $\square$

Proof. 8 Let us compute (9) knowing that $\|x\| = \sqrt{\langle x, x \rangle}$ then

$$\begin{aligned}\frac{1}{4}(\|x + y\|^2 - \|x - y\|^2) &= \frac{1}{4}(\langle x + y, x + y \rangle - \langle x - y, x - y \rangle) \\ &= \frac{1}{4}(\langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle - \\ &\quad - (\langle x, x \rangle + \langle x, -y \rangle + \langle -y, x \rangle + \langle -y, -y \rangle)) \\ &= \frac{1}{4}(\langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle - \\ &\quad - \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle - \langle y, y \rangle) \\ &= \frac{1}{4}(2\langle x, y \rangle + 2\langle x, y \rangle) \\ &= \langle x, y \rangle\end{aligned}$$

Where we used that $\langle x, y \rangle = \langle y, x \rangle$ since we are in a real inner product space. $\square$

Proof. 9 Let us compute (10) knowing that we are in a complex inner product space, then

$$\begin{aligned}
\frac{1}{4}(\|x+y\|^2 - \|x-y\|^2) &= \frac{1}{4}(\langle x+y, x+y \rangle - \langle x-y, x-y \rangle) \\
&= \frac{1}{4}(2\langle x, y \rangle + 2\langle y, x \rangle) \\
&= \frac{1}{4}(2\langle x, y \rangle + 2\overline{\langle x, y \rangle}) \\
&= \frac{1}{2}(\operatorname{Re}\langle x, y \rangle + i\operatorname{Im}\langle x, y \rangle + \operatorname{Re}\langle x, y \rangle - i\operatorname{Im}\langle x, y \rangle) \\
&= \operatorname{Re}\langle x, y \rangle
\end{aligned}$$

Now, we compute the second equation as follows

$$\begin{aligned}
\frac{1}{4}(\|x+iy\|^2 - \|x-iy\|^2) &= \frac{1}{4}(\langle x+iy, x+iy \rangle - \langle x-iy, x-iy \rangle) \\
&= \frac{1}{4}(\langle x, x \rangle + \langle x, iy \rangle + \langle iy, x \rangle + \langle iy, iy \rangle - \\
&\quad - (\langle x, x \rangle + \langle x, -iy \rangle + \langle -iy, x \rangle + \langle -iy, -iy \rangle)) \\
&= \frac{1}{4}(\langle x, x \rangle - i\langle x, y \rangle + i\langle y, x \rangle + \langle y, y \rangle - \\
&\quad - \langle x, x \rangle - i\langle x, y \rangle + i\langle y, x \rangle - \langle y, y \rangle) \\
&= \frac{1}{2}(-i\langle x, y \rangle + i\langle y, x \rangle) \\
&= \frac{1}{2}(-i\langle x, y \rangle + i\overline{\langle x, y \rangle}) \\
&= \frac{1}{2}(-i\operatorname{Re}\langle x, y \rangle + \operatorname{Im}\langle x, y \rangle + i\operatorname{Re}\langle x, y \rangle + \operatorname{Im}\langle x, y \rangle) \\
&= \operatorname{Im}\langle x, y \rangle
\end{aligned}$$

□

Proof. 10 Let z_1 and z_2 be complex numbers and let $\langle z_1, z_2 \rangle = z_1 \overline{z_2}$ we want to prove that in fact this is a valid definition of an inner product. So if this expression defines an inner product, it must have the four properties of inner products. Then for (IP1) let z_3 be also a complex number

$$\langle z_1 + z_3, z_2 \rangle = (z_1 + z_3) \overline{z_2} = z_1 \overline{z_2} + z_3 \overline{z_2} = \langle z_1, z_2 \rangle + \langle z_3, z_2 \rangle$$

For (IP2) we see that

$$\langle \alpha z_1, z_2 \rangle = \alpha z_1 \overline{z_2} = \alpha \langle z_1, z_2 \rangle$$

For (IP3) we see that

$$\langle z_1, z_2 \rangle = z_1 \overline{z_2} = \overline{\overline{z_1} z_2} = \overline{\langle z_2, z_1 \rangle}$$

Finally for (IP4) we see that $\langle z_1, z_1 \rangle = z_1 \overline{z_1} \geq 0$. Also, let $\langle z_1, z_1 \rangle = 0$ then

$$z_1 \overline{z_1} = (x + iy)(x - iy) = x^2 + y^2 = 0$$

Then the only way for this equation to hold is that $x = y = 0$ and hence $z_1 = 0$. Now, if we suppose $z_1 = 0$ then $\langle z_1, z_1 \rangle = z_1 \overline{z_1} = 0$ and hence (IP4) is also satisfied.

Therefore $\langle z_1, z_1 \rangle = z_1 \overline{z_1}$ defines in fact an inner product.

Also, to have orthogonality it must happen that $\langle z_1, z_2 \rangle = z_1 \overline{z_2} = 0$. $\square$

Proof. 11 Let X be the vector space of all ordered pairs of complex numbers. The norm defined as $\|x\| = |\xi_1| + |\xi_2|$ cannot be obtained from an inner product. We prove this by showing that the norm does not satisfy the parallelogram equality. Let us take $x = (-1, 1)$ and $y = (-1, -1)$ then

$$2(\|x\|^2 + \|y\|^2) = 2((|-1| + |1|)^2 + (|-1| + |-1|)^2) = 16$$

But

$$\|x + y\|^2 + \|x - y\|^2 = (|-1 - 1| + |1 - 1|)^2 + (|-1 + 1| + |1 + 1|)^2 = 8$$

Therefore the parallelogram equality is not satisfied and hence the norm defined cannot be obtained from an inner product. $\square$

Proof. **13** The inner product defined in 3.1-5 for continuous functions states that

$$\langle x, y \rangle = \int_a^b x(t) \overline{y(t)} dt$$

Let us check that this definition satisfies (IP1) to (IP4). We assume $x(t)$ and $y(t)$ complex-valued functions.

(IP1):

$$\begin{aligned} \langle x + y, z \rangle &= \int_a^b (x(t) + y(t)) \overline{z(t)} dt \\ &= \int_a^b x(t) \overline{z(t)} dt + \int_a^b y(t) \overline{z(t)} dt \\ &= \langle x, z \rangle + \langle y, z \rangle \end{aligned}$$

(IP2):

$$\langle \alpha x, y \rangle = \int_a^b \alpha x(t) \overline{y(t)} dt = \alpha \int_a^b x(t) \overline{y(t)} dt = \alpha \langle x, y \rangle$$

(IP3):

$$\langle x, y \rangle = \int_a^b x(t) \overline{y(t)} dt = \int_a^b \overline{\overline{x(t)} y(t)} dt = \overline{\langle y, x \rangle}$$

(IP4):

$$\langle x, x \rangle = \int_a^b x(t) \overline{x(t)} dt = \int_a^b |x(t)|^2 dt \geq 0$$

Let now $\langle x, x \rangle = 0$ then

$$\int_a^b |x(t)|^2 dt = 0$$

Then must be that $x(t) = 0$ since $x(t)$ is continuous. Conversely, if we suppose that $x(t) = 0$ we get that

$$\langle x, x \rangle = \int_a^b 0 \cdot 0 dt = 0$$

Therefore the inner product defined in 3.1-5 satisfy the properties (IP1) to (IP4). $\square$

3.2 - Further Properties of Inner Product Spaces

Proof. **2** An example of subspace of l^2 is the set sequences with finitely many entries different from zero followed by zeroes i.e.

$$Y = \{(x_1, x_2, \dots, x_k, 0, 0, 0, \dots) : x_i \in \mathbb{R}, 0 \leq i \leq k\}$$

Suppose $x = (x_1, x_2, \dots, x_k, 0, 0, 0, \dots) \in Y$ then

$$\|x\|_2 = \left(\sum_{i=1}^{\infty} |x_i|^2 \right)^{1/2} = \left(\sum_{i=1}^k |x_i|^2 \right)^{1/2} < \infty$$

So $Y \subset l^2$.

On the other hand, let $x' = (x'_1, x'_2, \dots, x'_k, 0, 0, 0, \dots) \in Y$, and α, β be scalars then

$$\begin{aligned} \alpha x + \beta x' &= \alpha(x_1, x_2, \dots, x_k, 0, 0, 0, \dots) + \beta(x'_1, x'_2, \dots, x'_k, 0, 0, 0, \dots) \\ &= (\alpha x_1, \alpha x_2, \dots, \alpha x_k, 0, 0, 0, \dots) + (\beta x'_1, \beta x'_2, \dots, \beta x'_k, 0, 0, 0, \dots) \\ &= (\alpha x_1 + \beta x'_1, \alpha x_2 + \beta x'_2, \dots, \alpha x_k + \beta x'_k, 0, 0, 0, \dots) \end{aligned}$$

We see that $\alpha x_i + \beta x'_i \in \mathbb{R}$ since $\mathbb{R}$ is a vector space then Y is a subspace of l^2 . $\square$

Proof. 3 Let X be the inner product space consisting of the polynomial $x = 0$ and all real polynomials in t of degree not exceeding 2, considered for real $t \in [a, b]$ with inner product defined as

$$\langle x, y \rangle = \int_a^b x(t)y(t) dt$$

Let $(x_n) = (\alpha_n t^2 + \beta_n t + \gamma_n) \in X$ be a Cauchy sequence on X where $\alpha_n, \beta_n, \gamma_n \in \mathbb{R}$ then since this is a Cauchy sequence, given $\epsilon > 0$ we can find $N \in \mathbb{N}$ such that when $n, m > N$ we have that

$$\|x_n - x_m\| = \left(\int_a^b (x_n(t) - x_m(t))^2 dt \right)^{1/2} < \epsilon$$

Hence

$$\left(\int_a^b ((\alpha_n - \alpha_m)t^2 + (\beta_n - \beta_m)t + (\gamma_n - \gamma_m))^2 dt \right)^{1/2} < \epsilon$$

For this integral to tend to 0 over the fixed interval $[a, b]$ since t does not depend on n must happen that $(\alpha_n - \alpha_m) \rightarrow 0$, $(\beta_n - \beta_m) \rightarrow 0$ and $(\gamma_n - \gamma_m) \rightarrow 0$.

This implies that (α_n) , (β_n) and (γ_n) are Cauchy sequences of real numbers which we know they converge, let us say that they converge to α, β and γ respectively.

Now, let us define $x = \alpha t^2 + \beta t + \gamma$ then

$$\begin{aligned} \|x_n - x\| &= \left(\int_a^b (x_n(t) - x(t))^2 dt \right)^{1/2} \\ &= \left(\int_a^b ((\alpha_n - \alpha)t^2 + (\beta_n - \beta)t + (\gamma_n - \gamma))^2 dt \right)^{1/2} \end{aligned}$$

But we know that $\alpha_n \rightarrow \alpha$, $\beta_n \rightarrow \beta$ and $\gamma_n \rightarrow \gamma$ therefore we have that

$$\left(\int_a^b ((\alpha_n - \alpha)t^2 + (\beta_n - \beta)t + (\gamma_n - \gamma))^2 dt \right)^{1/2} \rightarrow 0$$

Or what is the same $x_n \rightarrow x$ and hence since x_n was a Cauchy sequence then X is complete.

Let now Y be all $x \in X$ such that $x(a) = 0$. Let $y, y' \in Y$ and let α, β be scalars. We know that when $t = a$ then $y(a) = y'(a) = 0$ then for $z(t) = \alpha y(t) + \beta y'(t)$ we have that $z(a) = \alpha y(a) + \beta y'(a) = 0 + 0 = 0$. Therefore Y is a subspace of X .

Finally, let Z be all $x \in X$ of degree 2. Let $z, z' \in Z$ and let λ, ρ be scalars. Then let us compute $\lambda z + \rho z'$ as follows

$$\lambda(\alpha t^2 + \beta t + \gamma) + \rho(\alpha' t^2 + \beta' t + \gamma') = (\lambda\alpha + \rho\alpha')t^2 + (\lambda\beta + \rho\beta')t + (\lambda\gamma + \rho\gamma')$$

Where $\alpha, \alpha', \beta, \beta', \gamma$ and γ' are not zero. Then we see that if $\lambda = \rho = 0$ we get that $\lambda z + \rho z' = 0$ but the polynomial $z = 0$ is not included in Z then Z is not a subspace of X . $\square$

Proof. 4 Let $y \perp x_n$ and $x_n \rightarrow x$ then by the Continuity of the Inner Product we have that

$$\langle x_n, y \rangle \rightarrow \langle x, y \rangle$$

This implies that

$$\|\langle x_n, y \rangle - \langle x, y \rangle\| \rightarrow 0$$

But we know that $\langle x_n, y \rangle = 0$ because $y \perp x_n$ hence

$$\|\langle x, y \rangle\| \rightarrow 0$$

And by the (N2) property for norms, we get that

$$\langle x, y \rangle = 0$$

Therefore $x \perp y$. $\square$

Proof. **5** Let a sequence (x_n) in an inner product space such that $\|x_n\| \rightarrow \|x\|$ and $\langle x_n, x \rangle \rightarrow \langle x, x \rangle$. Given that $\|x_n\| \rightarrow \|x\|$ then $\|x_n\|^2 \rightarrow \|x\|^2$ so given $\epsilon^2/3 > 0$ there is $N \in \mathbb{N}$ such that when $n \geq N$ we see that

$$|\|x_n\|^2 - \|x\|^2| = |\langle x_n, x_n \rangle - \langle x, x \rangle| < \epsilon^2/3$$

On the other hand, since $\langle x_n, x \rangle \rightarrow \langle x, x \rangle$ given $\epsilon^2/3 > 0$ there is $N' \in \mathbb{N}$ such that when $n \geq N'$ we get that

$$|\langle x_n, x \rangle - \langle x, x \rangle| = |\langle x_n - x, x \rangle| < \epsilon^2/3$$

Also, let us note that

$$\begin{aligned} \|x_n - x\|^2 &= |\langle x_n - x, x_n - x \rangle| \\ &= |\langle x_n - x, x_n \rangle - \langle x_n - x, x \rangle| \\ &\leq |\langle x_n - x, x_n \rangle| + |\langle x_n - x, x \rangle| \\ &\leq |\langle x_n, x_n \rangle - \langle x, x_n \rangle| + |\langle x_n - x, x \rangle| \\ &\leq |\langle x_n, x_n \rangle - \langle x, x \rangle + \langle x, x \rangle - \langle x, x_n \rangle| + |\langle x_n - x, x \rangle| \\ &\leq |\langle x_n, x_n \rangle - \langle x, x \rangle| + |\langle x, x - x_n \rangle| + |\langle x_n - x, x \rangle| \end{aligned}$$

Then if we take $M = \max(N, N')$ when $n \geq M$ we have that

$$\begin{aligned} \|x_n - x\|^2 &\leq |\langle x_n, x_n \rangle - \langle x, x \rangle| + |\langle x, x - x_n \rangle| + |\langle x_n - x, x \rangle| \\ &< \epsilon^2/3 + \epsilon^2/3 + \epsilon^2/3 = \epsilon^2 \end{aligned}$$

Therefore $\|x_n - x\| < \epsilon$ and hence $x_n \rightarrow x$. □

Proof. 9 Let V be the vector space of all continuous complex-valued functions on $J = [a, b]$. Let $X_1 = (V, \|\cdot\|_\infty)$ where $\|x\|_\infty = \max_{t \in J} |x(t)|$ and let $X_2 = (V, \|\cdot\|_2)$ where

$$\|x\|_2 = \sqrt{\langle x, x \rangle} \quad \langle x, y \rangle = \int_a^b x(t) \overline{y(t)} dt$$

Also, let us define $i : X_1 \rightarrow X_2$ be the identity mapping $x \rightarrow x$. Suppose we take $x, y \in V$ then we see that

$$\begin{aligned} \|x - y\|_2 &= \sqrt{\langle x - y, x - y \rangle} \\ &= \sqrt{\int_a^b (x(t) - y(t)) \overline{(x(t) - y(t))} dt} \\ &= \sqrt{\int_a^b |x(t) - y(t)|^2 dt} \end{aligned}$$

But also we have that

$$\sqrt{\int_a^b |x(t) - y(t)|^2 dt} \leq \sqrt{\int_a^b \max_{t \in J} |x(t) - y(t)|^2 dt} = \max_{t \in J} |x(t) - y(t)| \sqrt{b - a}$$

So given $\epsilon > 0$ if we take $\delta = \epsilon / \sqrt{b - a}$ such that

$$\|x - y\|_\infty = \max_{t \in J} |x(t) - y(t)| < \delta = \frac{\epsilon}{\sqrt{b - a}}$$

We get that

$$\|x - y\|_2 \leq \max_{t \in J} |x(t) - y(t)| \sqrt{b - a} < \epsilon$$

Therefore the identity mapping i is continuous. □

Proof. 10 Let $T : X \rightarrow X$ be a bounded linear operator on a complex inner product space X and let $\langle Tx, x \rangle = 0$ for all $x \in X$.
Let $x = \alpha u + v \in X$ for some $u, v \in X$ and $\alpha \in \mathbb{C}$ then

$$\begin{aligned}\langle Tx, x \rangle &= \langle \alpha Tu + Tv, \alpha u + v \rangle \\ &= \langle \alpha Tu, \alpha u + v \rangle + \langle Tv, \alpha u + v \rangle \\ &= \alpha \bar{\alpha} \langle Tu, u \rangle + \alpha \langle Tu, v \rangle + \bar{\alpha} \langle Tv, u \rangle + \langle Tv, v \rangle \\ &= \alpha \langle Tu, v \rangle + \bar{\alpha} \langle Tv, u \rangle\end{aligned}$$

Then taking $\alpha = 1$ we get that

$$\begin{aligned}\langle Tu, v \rangle + \langle Tv, u \rangle &= 0 \\ \langle Tu, v \rangle &= -\langle Tv, u \rangle\end{aligned}$$

But if we take $\alpha = i$ we have that $i\langle Tu, v \rangle - i\langle Tv, u \rangle = 0$ or

$$\begin{aligned}\langle Tu, v \rangle - \langle Tv, u \rangle &= 0 \\ \langle Tu, v \rangle &= \langle Tv, u \rangle\end{aligned}$$

This implies that $\langle Tu, v \rangle = 0$ for all $u, v \in X$ so in particular if we select $v = Tu$ then $\langle Tu, Tu \rangle = 0$ and hence by (IP4) we have that $Tu = 0$. Therefore $T = 0$.

Let us consider now the Euclidean plane, a real inner product space, and let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos \pi/2 & -\sin \pi/2 \\ \sin \pi/2 & \cos \pi/2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$

Then let us compute the following inner product

$$\left\langle T \begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} \right\rangle = \left\langle \begin{bmatrix} -y \\ x \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} \right\rangle = -yx + xy = 0$$

Therefore, we see that the inner product is 0 but $T \neq 0$ so above result does not hold for real inner product spaces. $\square$

3.3 - Orthogonal Complements and Direct Sums

Proof. 1 Let H be a Hilbert space, $M \subset H$ a convex subset, and (x_n) a sequence in M such that $\|x_n\| \rightarrow d$ where $d = \inf_{x \in M} \|x\|$.

Let us consider $\overline{M}$ the closure of M then $\overline{M}$ is complete because $\overline{M} \subset H$, also, $(x_n) \in \overline{M}$.

Let $n, m \in \mathbb{N}$ where $n \geq m$. We see that $\frac{1}{2}x_n + \frac{1}{2}x_m \in M$ because M is convex but also we have that

$$\left\| \frac{x_n + x_m}{2} \right\| \geq d = \inf_{x \in M} \|x\|$$

Hence

$$\|x_n + x_m\| \geq 2d$$

On the other hand, by the parallelogram equality we get that

$$\begin{aligned} \|x_n - x_m\|^2 &= -\|x_n + x_m\|^2 + 2(\|x_n\|^2 + \|x_m\|^2) \\ &\leq -(2d)^2 + 2(\|x_n\|^2 + \|x_m\|^2) \end{aligned}$$

But since $\|x_n\| \rightarrow d$ as $n, m \rightarrow \infty$ then $\|x_n - x_m\| \rightarrow 0$. Therefore (x_n) is Cauchy, but this also implies that (x_n) converges because $\overline{M}$ is a subset of a Hilbert space. $\square$